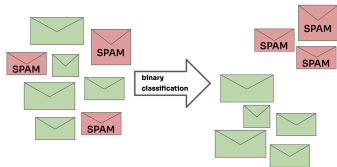
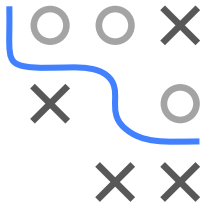


# Introduction to Machine Learning

## Classification Tasks

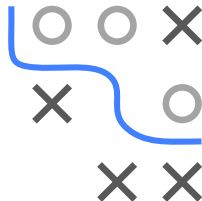


### Learning goals

- Classification is supervised learning with categorical labels
- Binary vs. multiclass
- Some examples of classification tasks

# CLASSIFICATION I

Learn functions that assign class labels to observation / feature vectors.  
Each observation belongs to exactly one class. The main difference to regression is that the target is categorical.



Our Data

| Sepal Length | Sepal Width | Petal Length | Petal Width | Species   |
|--------------|-------------|--------------|-------------|-----------|
| 5.1          | 3.5         | 1.4          | 0.2         | setosa    |
| 5.9          | 3.0         | 5.1          | 1.8         | virginica |



New Data with  
unknown label



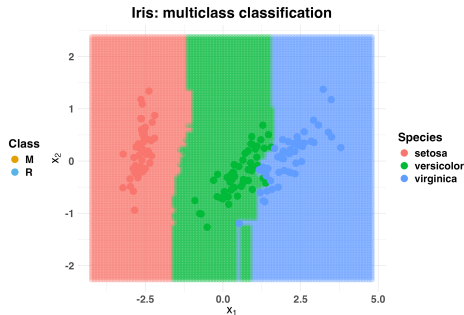
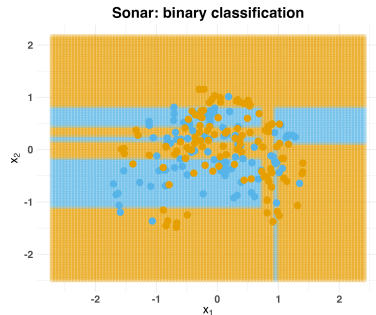
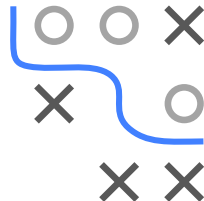
New Class label

| Sepal Length | Sepal Width | Petal Length | Petal Width | Species |
|--------------|-------------|--------------|-------------|---------|
| 5.4          | 3.3         | 3.2          | 1.1         | ???     |

# BINARY AND MULTICLASS TASKS I

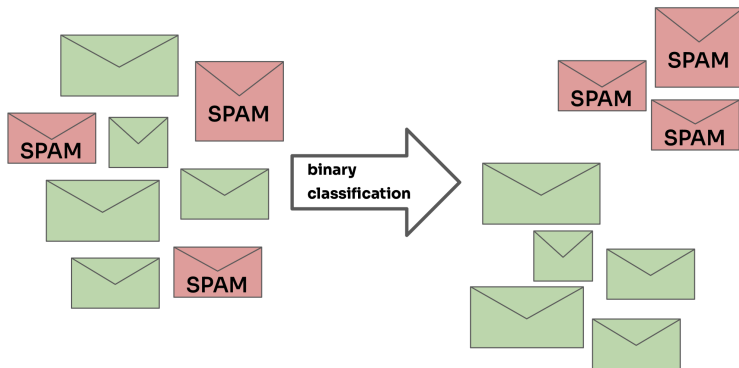
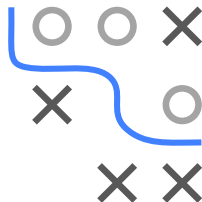
Tasks have a finite number of (unordered) classes.

They can be **binary** or **multiclass**.



# BINARY CLASSIFICATION TASK - EXAMPLES I

- Credit risk prediction, based on personal data and transactions
- Spam detection, based on textual features
- Churn prediction, based on customer behavior
- Predisposition for specific illness, based on genetic data



# MULTICLASS TASK - MEDICAL DIAGNOSIS I

INFO

SYMPTOMS

QUESTIONS

CONDITIONS

DETAILS

TREATMENT

Conditions that match your symptoms

UNDERSTANDING YOUR RESULTS ⓘ

Acute Sinusitis

error

Moderate match

>

Influenza (flu) adults

error

Moderate match

>

Common cold

error

Fair match

>

Asthma (Teen and Adult)

error

Fair match

>

Whooping cough

error

Fair match

>

↓

LOAD MORE CONDITIONS

Gender **Female**

Age **25**

Edit

My Symptoms

Edit

cough, headache, nausea

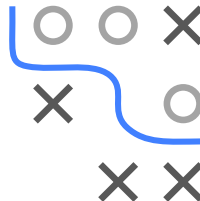
Could you be pregnant?

Edit

No

↺

Start Over



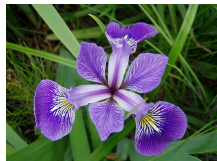
<https://symptoms.webmd.com>

# MULTICLASS TASK - IRIS I

The iris dataset was introduced by the statistician Ronald Fisher and is one of the most frequent used data sets. Originally, it was designed for linear discriminant analysis.



Setosa



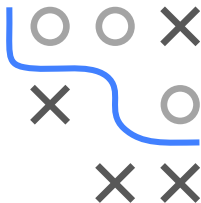
Versicolor



Virginica

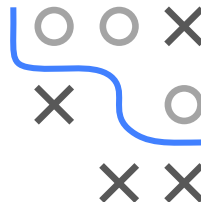
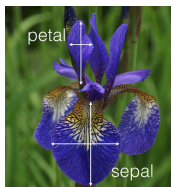
Source:

[https://en.wikipedia.org/wiki/Iris\\_flower\\_data\\_set](https://en.wikipedia.org/wiki/Iris_flower_data_set)



# MULTICLASS TASK - IRIS I

- 150 iris flowers
- Predict subspecies
- Based on sepal and petal length / width in [cm]



| ## |      | Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species   |
|----|------|--------------|-------------|--------------|-------------|-----------|
| ## | 1:   | 5.1          | 3.5         | 1.4          | 0.2         | setosa    |
| ## | 2:   | 4.9          | 3.0         | 1.4          | 0.2         | setosa    |
| ## | 3:   | 4.7          | 3.2         | 1.3          | 0.2         | setosa    |
| ## | 4:   | 4.6          | 3.1         | 1.5          | 0.2         | setosa    |
| ## | 5:   | 5.0          | 3.6         | 1.4          | 0.2         | setosa    |
| ## | ---  |              |             |              |             |           |
| ## | 146: | 6.7          | 3.0         | 5.2          | 2.3         | virginica |
| ## | 147: | 6.3          | 2.5         | 5.0          | 1.9         | virginica |
| ## | 148: | 6.5          | 3.0         | 5.2          | 2.0         | virginica |
| ## | 149: | 6.2          | 3.4         | 5.4          | 2.3         | virginica |
| ## | 150: | 5.9          | 3.0         | 5.1          | 1.8         | virginica |

# MULTICLASS TASK - IRIS I

